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Regression Analysis of Economic Factors Influencing Emigration Rate in Lithuania

Vilmante Kumpikaite^{a*}, Ineta Zickute

^a Kaunas University of Technology, Laisves av. 55, Kaunas 44209, Lithuania

Abstract

It becomes very important question how to stop emigration in Lithuania. The purpose of this paper is to present results of regression analysis modeling, representing the most important factors for emigration in Lithuania. Statistical data and regression analysis were used to achieve this purpose. Results revealed the regression equation with three major economic factors influencing emigration level: unemployment rate, Gini coefficient and Tax Freedom Day. This provided equation explains 70.7 percent of all indicators, influencing emigration rate. Following suggestions were provided in order to minimize emigration rate in Lithuania: 1) Unemployment rate should be reduced at least to 8.5 percent; 2) the level of Gini coefficient should be decreased to 30; and 3) tax paying time should be reduced by 5 working days.

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Keywords: Emigration; factors influencing emigration rate; unemployment rate; Gini coefficient; Tax Freedom Day.

1. Introduction

Citizens of higher developed countries are keenly aware of the impact of immigrants from less developed countries on their economy. Poor countries and poor people not only have less capital than rich ones but also have less knowledge. Above mentioned factors, such as still going transformation processes, low wages begun to stimulate citizens of these countries to migrate to the West European countries and the USA and to look for the better paid work and often for the better life. According to the 2004 Labor Force Survey, 10 percent of the British working-age population was born in another country (Dustmann & Fabbri, 2005).

* Corresponding author. Tel.: +370-37-323683; fax: +370-37-323683.

E-mail address: vilmante.kumpikaite@ktu.lt

Lithuania is a small Eastern European country with population of 3 millions. It became a member of EU from 2004. We should mention that Lithuania is a leader of emigrants' number for country in EU. The emigration rate is huge in Lithuania. It lost around 0.6 millions of population during period 1990-2010 (Europos migracijos tinklas [EMT], 2012). Immigration rate is too low to compensate losses of emigration. Therefore it becomes very important question how to stop emigration.

The purpose of this paper is to present the most important factors for emigration in Lithuania, using regression analysis.

2. Theoretical background

The factors that lead individuals to migrate, in turn, are motivated and sustained by three major types of influences – demand-pull factors in the destination area, supply-push factors in the origin area, and network factors that link origin and destination (Martin, 2003). There are as many reasons for migration as there are migrants, but most individuals who cross national borders do so for economic reasons.

Analyzed scientific literature on migration theories by Bauer and Zimmermann (1999), Berzinskiene (2006), Borjas (2008), Chiquiar and Hanson (2002), Kirstukas (2004), de Haas (2010), Haug (2008), Jennissen (2004), Kubursi (2006), Massey (1993), Wang (2010), Ciarniene and Kumpikaite (2011), Kumpikaite and Zickute (2012a) and others highlighted such factors influencing emigration: wage differences and income inequality, the economic development of country, politics of products' prices, conditions of trade, unequal division of labour force in sectors, establishment of new working places, unemployment rate, tax system, economic cycle, investments, and others. All groups of migration theories and factors influencing migration were analyzed and presented by Kumpikaite and Zickute (2012a, 2012b).

Emigration's indicator per 1000 population is the biggest in EU. This indicator was 23.7 per 1000 people in 2010, while the second place – 7.5 (Ireland), and third – 4.1 (Cyprus). Also negative net migration had Latvia, Bulgaria and Poland. Most countries have positive net migration and the highest indicator was 15.1 (Luxembourg).

During the first year of membership in European Union (2004) number of emigrants from Lithuania has grown by 1.4 times – from 22.7 thousands to 32.5 thousands. The largest number of persons – 48.1 thousand – emigrated in 2005. It was even twice more than in year 2003. During year 2006 and 2007 number of emigrants decreased. In 2007, 26.5 thousand persons left Lithuania to stay in foreign countries permanently or for a period longer than 6 months, which is by 1.3 thousand (4.7 percent) less than a year ago. Such situation depended on better economic and decreased unemployment rate in Lithuania.

Quite big problem is “brain drain”, as mostly migrants are working age, educated people. 77 percent of emigrants are ones of age group 15-44. A group of this age is just 43 percent of all Lithuanian population. Therefore we can see society's senescence which is a huge problem of most developed countries. It influences not just demographical problems but becomes disturbance for economics' growth as a country loses qualified and enterprising citizens (EMT, 2012).

3. Methodology

Regression analysis was used to depict the most important factors influencing emigration rate. Analysis was done using SPSS program and analyzing different indicators such as unemployment rate, unemployment duration, minimum and average salary, poverty levels, inequality of incomes, and tax rate. Statistical data of 2001-2011 was used for this analysis. Lack of such data of earlier years defined selected period. However, this period is sufficient to provide reliability of correlation and regression analysis.

Statistical significant model of multiple regressions is given in equation 1. It involves such indicators as unemployment rate (UR) in percent, Gini coefficient (Gini) and Tax Freedom Day (TFD).

$$E = 5.32 \times 10^{-24} \times UR^{0.320} \times Gini^{10.307} \times TFD^{5.019} \quad (1)$$

This model shows strong relation between all indicators involved in the equation and emigration rate. It includes 70.7 percent of all indicators, influencing emigration rate's changes. When unemployment rate rises by 1 percent and Gini coefficient and Tax Freedom Day stay the same then migration rate increases by 0.32 percent. When Gini coefficient grows by 1 percent and other indicators do not change, then emigration rate rises by 10.307 percent. In the case, when Tax Freedom Day becomes longer by 1 percent, while fixed Gini coefficient and unemployment rate, emigration rate increases by 5.019 percent. The equation with standardized regression coefficients shows that the biggest influence for emigration rate from highlighted indicators has Gini coefficient.

4. Comparison analysis of factors influencing emigration

As Lithuania's migration rate is the highest in all European Union, countries to which Lithuanians migrate the most is selected for this analysis. These countries are following: United Kingdom, Ireland, Norway and Germany. The average EU27 is taken for comparison as well.

4.1. Unemployment

Based on previous analysis unemployment rate in percent and percent of population who are unemployed from 6 to 11 months are the main indicators of unemployment criteria which influence migration. However lack of the data for the second indicator does not let to make analysis based on it. Therefore just comparison by unemployment rate of presented countries is made.

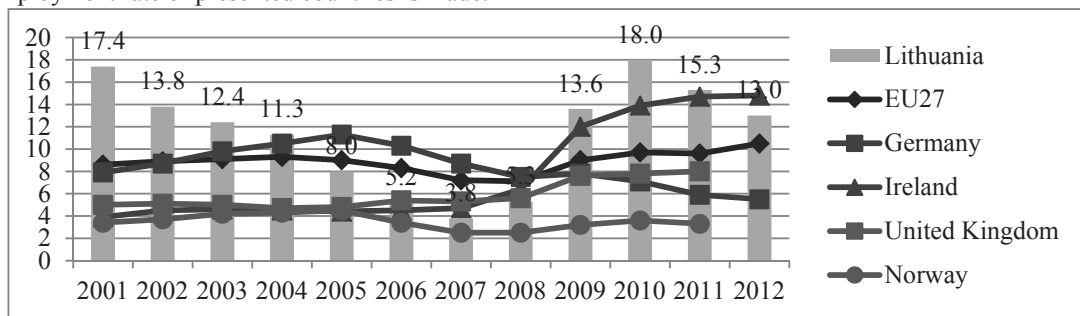


Fig. 1. Dynamics of unemployment rate in Lithuania, EU27 and countries to which Lithuanians migrate the most (the statistical office of the European Union Eurostat, 2013)

According to literature review normal level of unemployment is 5–6 percent, in critical case – 10 percent and more (Grazulis and Gruzevskis, 2008). As Figure 1 presents this normal unemployment level is just in Norway and Germany. The most amounts of countries went to the critical level of unemployment rate at the same time. Moreover, unemployment rate is the highest in Lithuania from analyzed countries and is higher by 2.5 percent from EU27 average and 2.2 times higher than normal unemployment rate.

Estimated regression equation of emigration and unemployment level showed that emigration rate started increasing when unemployment rate reached 8.5 percent.

4.2. Gini coefficient

Another very important problem is huge inequality of incomes in Lithuania. As we can see from Figure 2 EU27 average is 2.2 points lower than in Lithuania. Comparing Lithuania with countries where Lithuanians

migrate the most this indicator is one of the highest in Lithuania. As statistics and researches say, inequality of incomes becomes important issue when Gini coefficient is higher than 30.

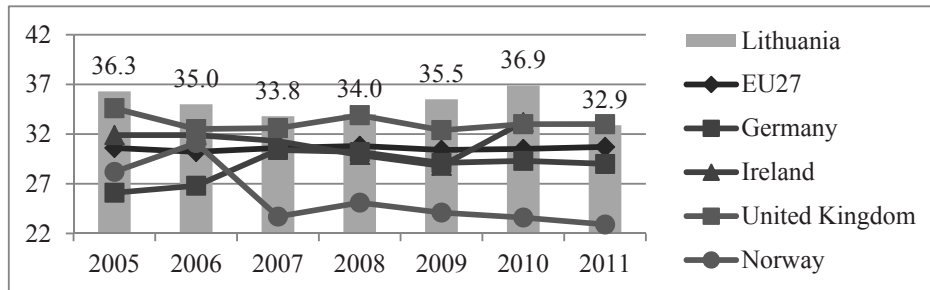


Fig. 2. Gini coefficient's dynamics in Lithuania, EU27 and countries to which Lithuanians migrate the most (the statistical office of the European Union Eurostat, 2013)

4.3. Tax Freedom Day

Based on Lithuanian free market Institutes colleagues from Belgium and France it is calculated how long European employee in average should work for taxes. Lithuania fits to top 10 European countries with the longest period according to this statistics. It was counted that Lithuanians had to work for taxes up to June 19th in 2011. It means 120 days. Looking at data in Figure 3, Lithuanians pay taxes 5 days more than EU27. United Kingdom, country with the biggest level of Lithuania's emigration, pays taxes just 97 days, Ireland – 92. However day of tax freedom is the highest in Germany and takes even 136 days.

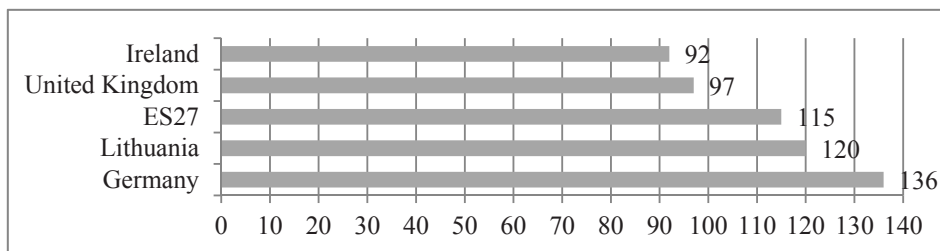


Fig. 3. Tax Freedom Day's dynamics in Lithuania and countries to which Lithuanians migrate the most in 2011 (Lithuanian Free Market Institute, 2012)

4.4. Modeling of Lithuanian's emigration forecast

If Lithuania could reach minimal unemployment rate up to 10 percent, to decrease Gini coefficient up to 30 and to reduce Tax Freedom Day down to 121, then amount of emigrants from Lithuania would be:

$$E = 4 \times (5.32 \times 10^{-24} \times 10^{0.320} \times 30^{10.307} \times 121^{5.019}) = \mathbf{2119} \quad (2)$$

However if indicators stay similar as previous years, then a number of Lithuanian emigrants would be:

$$E = 4 \times (5.32 \times 10^{-24} \times 15.4^{0.320} \times 37.6^{10.307} \times 126^{5.019}) = \mathbf{30563} \quad (3)$$

Looking at present situation it looks more likely that number of emigrants will reach 30563. However we know that usually we do not have either positive or negative variant. Therefore the most likely real situation was tried to count and according to it a number of emigrants per year would be about **8843**. Trying to reach this result

it is necessary to reduce unemployment rate by 2.7 percent, inequality of incomes by 3.8 points and freedom from taxes by 2 days. In this case Lithuania would become a bit closer to average of EU 27.

5. Conclusion

According to provided regression analysis using presented equation it was found that unemployment rate, Gini coefficient and Tax Freedom Day explains 70.7 percent of emigration reasons. Therefore, in order to decrease emigration rate in Lithuania unemployment rate should go down at least to 8.5 percent, the level of Gini coefficient decrease to 30.

One of suggestions for Lithuania tax system trying to decrease migration rate, it would be to reduce tax paying time by 5 working days and to reach an average of EU27. Speaking about taxation issues it is possibility to think about VAT decreasing as well. Regression analysis showed when VAT increases more than 18.5 percent a number of emigrants start growing rapidly. VAT is 21 percent at the moment in Lithuania. It would be useful to return it to 19 percent as it was before January of 2009. It would help people to survive taxes and increase purchasing power.

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